



Faculty of Computer Science and Mathematics
Universiti Malaysia Terengganu

LAB REPORT

CSE3023 Web-Based Application Development

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Lab Report	8
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Exercise:

Database

Table CarPricelist

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 Car_id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
☐	2 Brand	varchar(15)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	3 Model	varchar(30)	utf8mb4_general_ci		Yes	NULL			Change Drop More
☐	4 Cyclinder	int(11)			Yes	NULL			Change Drop More
☐	5 Price	double			Yes	NULL			Change Drop More

Source Code:

CarDAO.html

```
package com.DAO;

/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
/**
 *
 * @author wpy92
 */
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;

import com.Model.Car;

public class CarDAO {

    Connection connection = null;
    private String jdbcURL = "jdbc:mysql://localhost:3307/carshop";
    private String jdbcUsername = "root";
    private String jdbcPassword = "";
    private static final String INSERT_CAR_SQL = "INSERT INTO CarPricelist (Brand, Model, Cyclinder, Price) VALUES (?, ?, ?, ?)";
    private static final String SELECT_CAR_BY_ID = "SELECT * FROM CarPricelist WHERE Car_id = ?";
    private static final String SELECT_ALL_CARS = "SELECT * FROM CarPricelist";
    private static final String DELETE_CAR_SQL = "DELETE FROM CarPricelist WHERE Car_id=?";
    private static final String UPDATE_CAR_SQL = "UPDATE CarPricelist SET Brand = ?, Model = ?, Cyclinder = ?, Price = ? WHERE Car_id = ?";
```

```

public CarDAO() {
}

protected Connection getConnection() {
    Connection connection = null;
    try {
        Class.forName("com.mysql.jdbc.Driver");
        connection = DriverManager.getConnection(jdbcURL, jdbcUsername, jdbcPassword);
    }

    catch (SQLException e) {
        e.printStackTrace();
    }

    catch (ClassNotFoundException e) {
        e.printStackTrace();
    }

    return connection;
}

public void insertCar(Car car) throws SQLException {
    System.out.println(INSERT_CAR_SQL);
    // try-with-resource statement will auto close the connection.
    try (Connection connection = getConnection(); PreparedStatement preparedStatement = connection.prepareStatement(INSERT_CAR_SQL)) {
        preparedStatement.setString(1, car.getBrand());
        preparedStatement.setString(2, car.getModel());
        preparedStatement.setInt(3, car.getCylinder());
        preparedStatement.setDouble(4, car.getPrice());
        System.out.println(preparedStatement);
        preparedStatement.executeUpdate();
    }

    catch (SQLException e) {
        printSQLException(e);
    }
}

```

```

public Car selectCar(int id) {
    Car car = null;
    // Step 1: Establishing a Connection
    try (Connection connection = getConnection();

        // Step 2: Create a statement using connection object
        PreparedStatement preparedStatement = connection.prepareStatement(SELECT_CAR_BY_ID);) {
        preparedStatement.setInt(1, id);

        System.out.println(preparedStatement);
        // Step 3: Execute the query or update query
        ResultSet rs = preparedStatement.executeQuery();

        // Step 4: Process the ResultSet object.
        while (rs.next()) {
            String brand = rs.getString("Brand");
            String model = rs.getString("Model");
            int cylinder = rs.getInt("Cylinder");
            double price = rs.getDouble("Price");
            car = new Car(id, brand, model, cylinder, price);
        }
    } catch (SQLException e) {
        printSQLException(e);
    }

    return car;
}

```

```

public List<Car> selectAllCars() {
    // using try-with-resources to avoid closing resources (boiler plate code)
    List<Car> cars = new ArrayList<>();
    // Step 1: Establishing a Connection
    try (Connection connection = getConnection();
        // Step 2: Create a statement using connection object
        PreparedStatement preparedStatement = connection.prepareStatement(SELECT_ALL_CARS);) {
        System.out.println(preparedStatement);
        // Step 3: Execute the query or update query
        ResultSet rs = preparedStatement.executeQuery();

        // Step 4: Process the ResultSet object.
        while (rs.next()) {
            int id = rs.getInt("Car_id");
            String brand = rs.getString("Brand");
            String model = rs.getString("Model");
            int cylinder = rs.getInt("Cylinder");
            double price = rs.getDouble("Price");
            cars.add(new Car(id, brand, model, cylinder, price));
        }
    }

    catch (SQLException e) {
        printSQLException(e);
    }

    return cars;
}

public boolean deleteCar(int id) throws SQLException {
    boolean rowDeleted;
    try (Connection connection = getConnection(); PreparedStatement statement = connection.prepareStatement(DELETE_CAR_SQL);) {
        statement.setInt(1, id);
        rowDeleted = statement.executeUpdate() > 0;
    }
    return rowDeleted;
}

public boolean updateCar(Car car) throws SQLException {
    boolean rowUpdated;
    try (Connection connection = getConnection(); PreparedStatement statement = connection.prepareStatement(UPDATE_CAR_SQL);) {
        statement.setString(1, car.getBrand());
        statement.setString(2, car.getModel());
        statement.setInt(3, car.getCylinder());
        statement.setDouble(4, car.getPrice());
        statement.setInt(5, car.getCar_id());
        rowUpdated = statement.executeUpdate() > 0;
    }
    return rowUpdated;
}

private void printSQLException(SQLException ex) {
    for (Throwable e : ex) {
        if (e instanceof SQLException) {
            e.printStackTrace(System.err);
            System.err.println("SQLState: " + ((SQLException) e).getSQLState());
            System.err.println("Error Code: " + ((SQLException) e).getErrorCode());
            System.err.println("Message: " + e.getMessage());
            Throwable t = ex.getCause();
            while (t != null) {
                System.out.println("Cause: " + t);
                t = t.getCause();
            }
        }
    }
}
}

```

Car.java

```
package com.Model;

/**
 *
 * @author wpy92
 */
public class Car {

    //Define instant variables
    protected int car_id;
    protected String brand;
    protected String model;
    protected int cylinder;
    protected double price;

    //Constructors
    public Car() {
    }

    public Car(String brand, String model,
               int cylinder, double price) {

        this.brand = brand;
        this.model = model;
        this.cylinder = cylinder;
        this.price = price;
    }

    public Car(int car_id, String brand,
               String model, int cylinder,
               double price) {

        this.car_id = car_id;
        this.brand = brand;
        this.model = model;
        this.cylinder = cylinder;
        this.price = price;
    }

    //Getters and Setters
    public int getCar_id() {
        return car_id;
    }

    public void setCar_id(int car_id) {
        this.car_id = car_id;
    }

    public String getBrand() {
        return brand;
    }

    public void setBrand(String brand) {
        this.brand = brand;
    }

    public String getModel() {
        return model;
    }

    public void setModel(String model) {
        this.model = model;
    }

    public int getCylinder() {
        return cylinder;
    }

    public void setCylinder(int cylinder) {
        this.cylinder = cylinder;
    }

    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }
}
```

CarServlet.java

```
package com.WEB;

import java.io.IOException;
import java.sql.SQLException;
import java.util.List;
import java.io.PrintWriter;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

import com.DAO.CarDAO;
import com.Model.Car;

/**
 *
 * @author wpy92
 */
@WebServlet(urlPatterns = {
    "/list",
    "/new",
    "/insert",
    "/edit",
    "/update",
    "/delete"
})
public class CarServlet extends HttpServlet {

    private CarDAO carDAO;

    public void init() {
        carDAO = new CarDAO();
    }

    /**
     * Processes requests for both HTTP GET and POST
     * methods.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            /* TODO output your page here. You may use following sample code. */
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet EmployeeServlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Servlet EmployeeServlet at " + request.getContextPath() + "</h1>");
            out.println("</body>");
            out.println("</html>");
        }
    }
}
```

```

// <editor-fold defaultstate="collapsed" desc="HttpServlet methods. Click on the + sign on the left to edit the code.">
/**
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    String action = request.getServletPath();
    try {
        switch (action) {
            case "/new":
                showNewForm(request, response);
                break;

            case "/insert":
                insertCar(request, response);
                break;

            case "/delete":
                deleteCar(request, response);
                break;

            case "/edit":
                showEditForm(request, response);
                break;

            case "/update":
                updateCar(request, response);
                break;

            default:
                listCars(request, response);
                break;
        }
    } catch (SQLException ex) {
        throw new ServletException(ex);
    }
}

/**
 * Handles the HTTP <code>POST</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    doGet(request, response);
}

private void listCars(HttpServletRequest request,
    HttpServletResponse response)
    throws SQLException, IOException, ServletException {
    List<Car> listCars = carDAO.selectAllCars();
    request.setAttribute("listCars", listCars);
    RequestDispatcher dispatcher = request.getRequestDispatcher("carList.jsp");
    dispatcher.forward(request, response);
}

```

```

private void showNewForm(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {

    RequestDispatcher dispatcher = request.getRequestDispatcher("carForm.jsp");

    dispatcher.forward(request, response);
}

private void showEditForm(HttpServletRequest request,
    HttpServletResponse response)
    throws SQLException, ServletException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));

    Car existingCar = carDAO.selectCar(id);

    RequestDispatcher dispatcher = request.getRequestDispatcher("carForm.jsp");

    request.setAttribute("car", existingCar);

    dispatcher.forward(request, response);
}

private void insertCar(HttpServletRequest request,
    HttpServletResponse response)
    throws SQLException, IOException {

    String brand = request.getParameter("brand");
    String model = request.getParameter("model");
    int cylcylinder = Integer.parseInt(request.getParameter("cylcylinder"));
    double price = Double.parseDouble(request.getParameter("price"));
    Car newCar = new Car(brand, model, cylcylinder, price);
    carDAO.insertCar(newCar);
    response.sendRedirect("list");
}

private void updateCar(HttpServletRequest request,
    HttpServletResponse response)
    throws SQLException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));
    String brand = request.getParameter("brand");
    String model = request.getParameter("model");
    int cylcylinder = Integer.parseInt(request.getParameter("cylcylinder"));
    double price = Double.parseDouble(request.getParameter("price"));

    Car car = new Car(id, brand, model, cylcylinder, price);
    carDAO.updateCar(car);

    response.sendRedirect("list");
}

private void deleteCar(HttpServletRequest request,
    HttpServletResponse response)
    throws SQLException, IOException {

    int id = Integer.parseInt(request.getParameter("id"));

    carDAO.deleteCar(id);

    response.sendRedirect("list");
}
}

```

web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app version="4.0" xmlns="http://xmlns.jcp.org/xml/ns/javaee" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocati
    <servlet>
        <servlet-name>CarServlet</servlet-name>
        <servlet-class>com.WEB.CarServlet</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>CarServlet</servlet-name>
        <url-pattern>/CarServlet</url-pattern>
    </servlet-mapping>

    <!-- Ensure CSS, JavaScript and image files are served correctly
    when using servlet mapping "/" -->
    <servlet-mapping>
        <servlet-name>default</servlet-name>
        <url-pattern>*.css</url-pattern>
    </servlet-mapping>

    <servlet-mapping>
        <servlet-name>default</servlet-name>
        <url-pattern>*.js</url-pattern>
    </servlet-mapping>

    <servlet-mapping>
        <servlet-name>default</servlet-name>
        <url-pattern>*.png</url-pattern>
    </servlet-mapping>

    <session-config>
        <session-timeout>
            30
        </session-timeout>
    </session-config>
</web-app>
```

carForm.jsp

```
<!--
Document   : carForm.jsp
Created on : Jun 2, 2026, 9:33:49 PM
Author    : wpy92
-->

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<!DOCTYPE html>
<html>
<head>
<title>Car Shop Management</title>

<link rel="stylesheet"
href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css"
integrity="sha384-ggOyR0iXCbMQv3Xipma34MD+dH/1fQ784/j6cY/iJTQUOhcWr7x9JvoRxT2MZw1T"
crossorigin="anonymous">

</head>
<body>
<header>
<nav class="navbar navbar-expand-md navbar-dark"
style="background-color: steelblue">

<div>
<a href="" class="navbar-brand">
Car Shop Management
</a>
</div>

<ul class="navbar-nav">
<li>
<a href="<%=request.getContextPath()%>/list"
class="nav-link">Cars</a>
</li>
</ul>

</nav>
</header>

<br>
```

```

<div class="container col-md-5">
  <div class="card">
    <div class="card-body">
      <:if test="{car != null}">
        <form action="update" method="post">
        </:if>

      <:if test="{car == null}">
        <form action="insert" method="post">
        </:if>

      <h2>
        <:if test="{car != null}">
          Edit Car
        </:if>

        <:if test="{car == null}">
          Add New Car
        </:if>

      </h2>

      <:if test="{car != null}">
        <input type="hidden"
          name="id"
          value="<:out value='{car.car_id}' />" />
      </:if>

      <fieldset class="form-group">
        <label>Brand</label>

        <input type="text"
          class="form-control"
          name="brand"
          value="<:out value='{car.brand}' />"
          required="required">

      </fieldset>

```

```

<fieldset class="form-group">

    <label>Model</label>

    <input type="text"
           class="form-control"
           name="model"
           value="<c:out value='${car.model}' />"
           required="required">

</fieldset>

<fieldset class="form-group">

    <label>Cyclinder</label>

    <input type="number"
           class="form-control"
           name="cyclinder"
           value="<c:out value='${car.cyclinder}' />"
           required="required">

</fieldset>

<fieldset class="form-group">

    <label>Price (RM)</label>

    <input type="number"
           step="0.01"
           class="form-control"
           name="price"
           value="<c:out value='${car.price}' />"
           required="required">

</fieldset>

<button type="submit" class="btn btn-success">
    Save
</button>

</form>

</div>

</div>

</div>

</body>
</html>

```

carList.jsp

```
<%--
Document      : carList
Created on    : Jun 2, 2026, 9:36:24 PM
Author       : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@ taglib uri="http://java.sun.com/jsp/jstl/core" prefix="c"%>
<!DOCTYPE html>
<html>
  <head>
    <title>Car Shop Management</title>

    <link rel="stylesheet"
          href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css"
          integrity="sha384-ggOyR0iXCbMQV3Xipma34MD+dH/1fQ784/j6cY/iJTQUOhcWr7x9JvoRxT2MZw1T"
          crossorigin="anonymous">

  </head>
  <body>
    <header>
      <nav class="navbar navbar-expand-md navbar-dark"
           style="background-color: steelblue">

        <div>
          <a href="" class="navbar-brand">
            Car Shop Management
          </a>
        </div>

        <ul class="navbar-nav">
          <li>
            <a href="<%=request.getContextPath()%>/list" class="nav-link">Cars</a>
          </li>
        </ul>

      </nav>
    </header>

    <br>
```


index.jsp

```
<%--
Document   : index
Created on : Jun 2, 2026, 9:42:47 PM
Author    : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
  <head>
    <title>Car Shop MVC Application</title>
    <link rel="stylesheet"
          href="https://stackpath.bootstrapcdn.com/bootstrap/4.3.1/css/bootstrap.min.css"
          integrity="sha384-ggOyR0iXCbMQv3Xipma34MD+dH/1fQ784/j6cY/iJTQUOhcWr7x9JvoRxT2MZw1T"
          crossorigin="anonymous">
    </head>
    <body>
      <h1>Car Shop MVC System</h1>
      <br>
      <ul>
        <li>
          <a href="http://localhost:8080/CarPricelist/list">
            View All Cars
          </a>
        </li>
        <li>
          <a href="http://localhost:8080/CarPricelist/new">
            Add New Car
          </a>
        </li>
        <li>
          <a href="http://localhost:8080/CarPricelist/list">
            Edit Car
          </a>
        </li>
      </ul>
    </body>
  </html>
```

error.jsp

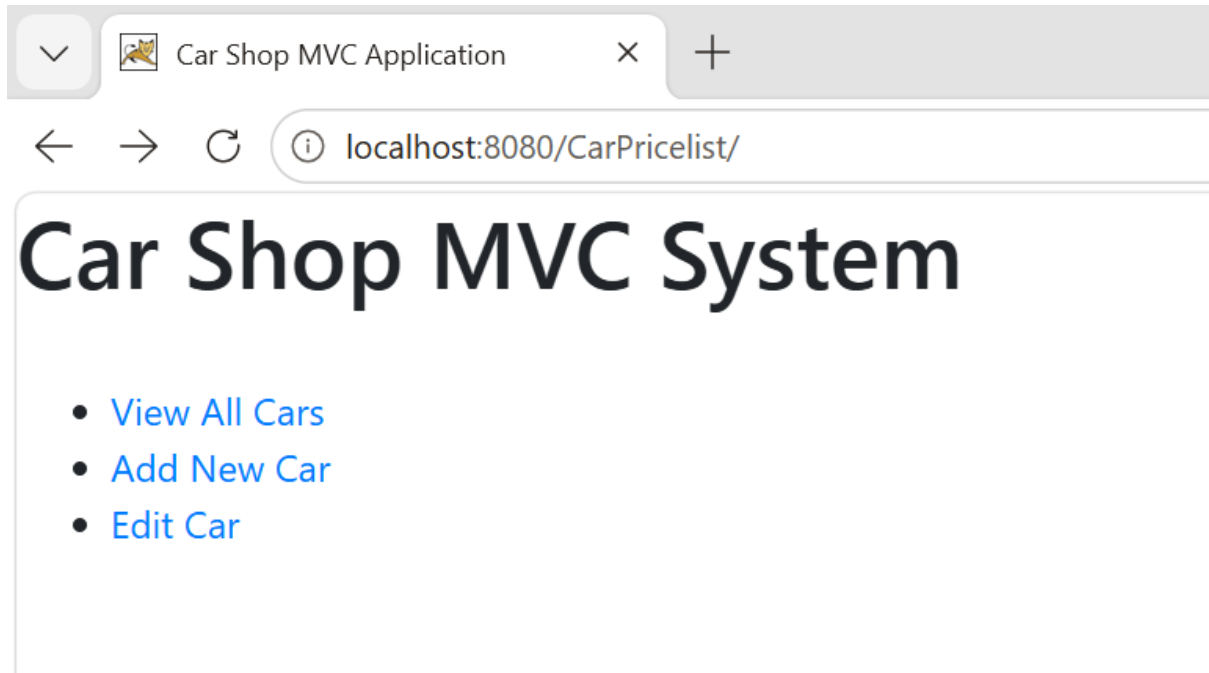
```
<%--
    Document      : error
    Created on    : Jun 2, 2026, 9:40:51 PM
    Author       : wpy92
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">

<html>
<head>
<title>Error page</title>
</head>
<body>
<center>
<h1>Error</h1>
<h2><%=exception.getMessage()%><br/> </h2>
</center>
</body>
</html>
```

Output:

index.jsp



CRUD (include carForm.jsp and carList.jsp)

Create/ Add new data

Car Shop Management

localhost:8080/CarPricelist/new

Car Shop Management Cars

Add New Car

Brand

Proton

Model

Saga

Cyclinder

4

Price (RM)

37990

Save

↔T↔

▼

Car_id

Brand

Model

Cyclinder

Price

☐

 Edit

 Copy

 Delete

1

Perodua

Axia

3

22000

☐

 Edit

 Copy

 Delete

2

Proton

Saga

4

37990

View/ View all data

Car Shop Management

localhost8080/CarPricelist/list

Car Shop Management Cars

Car Price List

Add New Car

ID	Brand	Model	Cylinder	Price (RM)	Actions
1	Perodua	Axia	3	22000.0	Edit Delete
2	Proton	Saga	4	37990.0	Edit Delete

← T →

▼ Car_id Brand Model Cylinder Price

☐  Edit  Copy  Delete 1 Perodua Axia 3 22000

☐  Edit  Copy  Delete 2 Proton Saga 4 37990

Update/ Edit current data

Before

Car Shop Management Cars

Edit Car

Brand
Proton

Model
Saga

Cylinder
4

Price (RM)
37990.0

Save

	Car_id	Brand	Model	Cylinder	Price
☐	1	Perodua	Axia	3	22000
☐	2	Proton	Saga	4	37990

After

Car Shop Management Cars

Edit Car

Brand
Proton

Model
S70

Cylinder
3

Price (RM)
68880

Save

Car Shop Management

localhost:8080/CarPricelist/list

Car Shop Management Cars

Car Price List

Add New Car

ID	Brand	Model	Cylinder	Price (RM)	Actions
1	Perodua	Axia	3	22000.0	Edit Delete
2	Proton	S70	3	68880.0	Edit Delete

← T →

▼

	Car_id	Brand	Model	Cylinder	Price
☐ Edit Copy Delete	1	Perodua	Axia	3	22000
☐ Edit Copy Delete	2	Proton	S70	3	68880

Delete/ Delete the specific data

Before

Car Shop Management Cars

Car Price List

Add New Car

ID	Brand	Model	Cylinder	Price (RM)	Actions
1	Perodua	Axia	3	22000.0	Edit Delete
2	Proton	S70	3	68880.0	Edit Delete

Car Shop Management Cars

	Car_id	Brand	Model	Cylinder	Price
☐		Edit	Copy	Delete	1 Perodua Axia 3 22000
☐		Edit	Copy	Delete	2 Proton S70 3 68880

After

Car Shop Management Cars

Car Price List

Add New Car

ID	Brand	Model	Cylinder	Price (RM)	Actions
1	Perodua	Axia	3	22000.0	Edit Delete

Car Shop Management Cars

	Car_id	Brand	Model	Cylinder	Price
☐		Edit	Copy	Delete	1 Perodua Axia 3 22000